

# BEARINGS & THEIR NUMBERS

The precision part behind every machine — how to read a bearing code, choose a type, size it, fit it and keep it alive, all in one sheet.

|                                                                                           |                                               |
|-------------------------------------------------------------------------------------------|-----------------------------------------------|
| Ball · Roller · Plain                                                                     | three great bearing families, dozens of types |
| up to 90°                                                                                 | contact angle — in angular-contact bearings   |
| P0 → P6                                                                                   | ISO precision classes; P6 finest              |
| 10  revs | life rated in millions of revolutions         |
| 100s of kN                                                                                | load ratings from ~0.5 kN to 100s of kN       |

## SECTION 2 · THE BASICS

### WHAT IS A BEARING?

A bearing is a machine element that supports a rotating (or sliding) shaft, holding it in position while it spins and carrying the loads that push on it. Without it, a shaft running directly in a hole would grind, wear and seize within minutes.

A bearing does three jobs in one part:

- Supports the shaft — locates it radially and axially in one spot
- Carries the load — radial (across the shaft) and axial (along it)
- Reduces friction — rolling or low-friction sliding instead of metal-on-metal

In a **rolling bearing** the moving parts roll — balls or rollers between two hardened steel rings — so sliding friction is replaced by much lower rolling friction. In a **plain bearing** the shaft slides in a soft, self-lubricating sleeve. Every motor, gearbox, wheel, pump, fan, spindle and conveyor on Earth leans on one of these two ideas.

It is the 'precision part' of the machine: raceways are ground and lapped to microns, balls are accurate to a few micrometres, and the whole package costs pocket change — yet a single seized bearing can stop an entire production line, a washing machine or a Formula 1 car.

## SECTION 3 · FAMILIES

### BEARING TYPES — LOADS & USES

| Type                            | Loads                     | Where it shines                                    |
|---------------------------------|---------------------------|----------------------------------------------------|
| <b>Deep-groove ball (6)</b>     | Radial + some axial       | The workhorse: motors, fans, wheels, gearboxes     |
| <b>Angular-contact (7)</b>      | Combined, one direction   | High-speed spindles, machine tools — used in pairs |
| <b>Cylindrical roller (NNU)</b> | Heavy radial              | Gearboxes, electric motors, rolling mills          |
| <b>Tapered roller (3)</b>       | Heavy radial + axial      | Car wheel hubs (a classic pair), axes              |
| <b>Spherical roller (2)</b>     | Very heavy, self-aligning | Mining, conveyors, paper mills, big shafts         |
| <b>Needle roller</b>            | Radial in tight space     | Thin, strong: gearboxes, rocker arms, U-joints     |
| <b>Thrust ball (5)</b>          | Axial only                | Shaft ends, turntables, screw jacks                |
| <b>Ball thrust</b>              | Axial, low–medium speed   | Pulleys, turntables, machine-tool spindles         |
| <b>Plain / sleeve</b>           | Radial (sliding)          | Crankshafts, electric motors, hinges, bushes       |

Rule of thumb: **ball** bearings for speed and moderate load, **roller** bearings for heavy load, **thrust** for axial load alone, **plain** for cheaps, low-speed sliding or oscillating motion.

## SECTION 4 · THE CODE

### ROLLING BEARING NUMBERS (ISO 15 / DIN 623)

The code printed on a bearing describes its **type**, **size series** and **bore** in a short string — usually read as four groups:

**6205-2RS** → 6 | 2 | 05 | -2RS

- 1. TYPE digit** — the bearing family. 6 = deep-groove ball, 7 = angular-contact, 3 = tapered roller, 2 = spherical roller, 5 = thrust, NNU = cylindrical roller.
- 2. SERIES digits** — the width/strength class. 60 = extra light, 62 = light, 63 = medium, 64 = heavy. Bigger rings + bigger balls = stronger for the same shaft.
- 3. BORE code** — last two digits × 5 mm = the shaft (bore) size. Special codes below 04.
- 4. SUFFIX** — letters for seals, shields and clearance. Anything after a hyphen or letter is an option, not the size.

Three worked examples — decode them in 10 seconds:

- 6204 = deep-groove ball · light series · bore 04×5 = 20 mm
- 6305 = deep-groove ball · medium series · bore 05×5 = 25 mm
- 6201-2RS = deep-groove · light · bore 01 = 12 mm · no rubber seals (-2RS)

So the code tells you everything before you reach for a calliper: the family on the first digit, the strength on the middle digits, the bore on the last two, and the sealing on the suffix.

## SECTION 5 · READING TABLES

### HOW TO READ A BEARING NUMBER TABLE

Catalogue tables give, for every code: **d** = **bore**, **D** = **outside diameter**, **B** = **width**, and the load ratings. The bore is the number that must match your shaft.

**THE BORE RULE** — last two digits × 5 mm gives the bore for codes 04 and up. The special small sizes are memorised, not multiplied:

00 = 10 mm · 01 = 12 mm · 02 = 15 mm · 03 = 17 mm · then 04 ≈ 20, 05 = 25, 06 = 30 ...

Below 10 mm bore, the whole number is the bore: 608 = 8 mm, 623 = 3 mm. From 04 up, just multiply.

Worked decoding of a few real codes:

| Code     | Type / series                 | Bore             | OD    | Width |
|----------|-------------------------------|------------------|-------|-------|
| 6204     | ball · light                  | 20 mm (04×5)     | 47 mm | 14 mm |
| 6305     | ball · medium                 | 25 mm (05×5)     | 62 mm | 17 mm |
| 6201-2RS | ball · light · 2 seals        | 12 mm (01)       | 32 mm | 10 mm |
| 608-2Z   | ball · 8 mm shaft · 2 shields | 8 mm (whole no.) | 22 mm | 7 mm  |
| 6308     | ball · medium                 | 40 mm (08×5)     | 90 mm | 23 mm |
| NU 205   | cylindrical roller            | 25 mm (05×5)     | 52 mm | 15 mm |
| 30205    | tapered roller                | 25 mm (05×5)     | 52 mm | 15 mm |

Same bore, bigger series = bigger, stronger bearing. A 6204 (20 mm) is much smaller and lighter than a 6304 (20 mm); the 63 series has fatter rings and bigger balls for a heavier load.

## SECTION 6 · SIZE & CLEARANCE

### BEARING DIMENSIONS & CLEARANCES

Three dimensions are on every bearing table: bore (d) grips the shaft, outside diameter (D) sits in the housing, width (B) is the ring thickness. Fit your shaft to d and your bore/housing to D.

**RADIAL CLEARANCE** — the tiny working gap between the balls and the raceways. It lets the bearing run, breathe with temperature, and align slightly.

| Code        | Clearance          | Use it when                          |
|-------------|--------------------|--------------------------------------|
| C2          | less than normal   | tight, precise running               |
| C0 / normal | standard (no code) | most general-purpose jobs            |
| C3          | more than normal   | heat, speed, press-fit on both rings |
| C4          | extra              | very hot / very fast running         |

A press (interference) fit squeezes the inner ring, taking up clearance — so a bearing force-fit onto a hot shaft often needs C3 to keep a working gap. The clearance is printed as a suffix, e.g. 6204-2RS C3.

**SEALING — open vs shielded vs sealed:**

- Open** — no protection; grease of your choice, needs re-lubricating
- Z / ZZ** — non-contact metal shields keep big dirt out but not fine dust
- 2RS / RS** — rubber (nitrile) seals, contact type, greased for life
- N / NR** — snap-ring groove / snap ring fitted so the bearing clicks into a housing bore

Sealed (2RS/ZZ) bearings are **greased for life** — never open them. Choose ZZ where friction and heat matter (fast spindles), 2RS where contamination rules (wheels, motors).

## SECTION 7 · BUILD

### MATERIALS & CONSTRUCTION

Rolling bearings are made from **high-carbon chrome bearing steel** — 52100 (EN31) in the West, **Gcr15** (China) in Asia. It hardens to ~60–67 HRC, giving raceways that outlast everything around them.

The parts of a rolling bearing:

- Inner ring** — presses onto the shaft; its raceway is the inner track
- Outer ring** — sits in the housing; its raceway is the outer track
- Rolling elements** — balls or rollers that roll between the races
- Cage (retainer)** — keeps the elements evenly spaced so they can't touch
- Seals / shields** — keep grease in and dirt out
- Grease or oil** — the film that keeps metal off metal

CAGES — the weak link, and they hate shock. **Steel** (stamped or machined) is strongest and cheapest; **brass** (machined) resists heat and runs smoother; **polymer** (nylon, PA66) is light, quiet, corrosion-proof and self-lubricating — the modern default for everyday ball bearings.

**LUBRICATION — grease vs oil:**

| Grease                          | Oil                                |
|---------------------------------|------------------------------------|
| Stays where you put it          | Flows &amp; carries heat away      |
| Sealed bearings = lifetime fill | Needed above ~dN 300 000           |
| Simpler, cleaner, cheaper       | Best at very high speed            |
| Fill only 30–50% of free space  | Level should reach the lowest ball |

Too much grease is as bad as too little — it **churns**, and **churns generate heat**. Open bearings: pack 30–50%. NLGI grade 2 (soft butter) suits almost every general bearing; NLGI 3 suits sealed motor bearings.

## SECTION 8 · LOADS & LIFE

### LOADS, RATINGS & THE L10 LIFE FORMULA

A bearing must carry two kinds of load: **radial** (at right angles to the shaft — the weight it supports) and **axial / thrust** (along the shaft — push or pull). Deep-groove ball bearings carry both; thrust bearings carry axial only. The more of either, the shorter the life.

| Rating                     | Symbol | What it means                                                                                                                                                                                                                        | How it is used                                           |
|----------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Dynamic load rating</b> | C      | Load a bearing can carry for 10  revolutions with 90% surviving                                                                                    | The numerator of the L10 life formula                    |
| <b>Static load rating</b>  | C0     | Max load without permanent denting (no rotation)                                                                                                                                                                                     | Sets the limit for shock loads &amp; stationary bearings |
| <b>Equivalent load</b>     | P      | Single value combining radial + axial effects (P = X·F  + Y·F  | The actual stress you feed into the life equation        |
| <b>Preload</b>             | —      | Forcing elements against the races before running, to kill play                                                                                                                                                                      | Spindles, robot joints, angular-contact pairs            |

**THE L10 LIFE FORMULA** — the ISO prediction for 90% of a batch surviving:

**L  = (C ÷ P)³ × 10  revolutions** for ball bearings · **L  = (C ÷ P)  × 10  for roller bearings**

Here **C** is the catalogue dynamic rating and **P** is your equivalent load. The exponent does the damage: double the load and a ball-bearing life drops by 2³ = 8×, a roller by ~10×. That cubic relationship is why a lightly-loaded bearing can outlive the machine, while an overloaded one dies in weeks. To read life in hours, divide revolutions by speed: **L  = 10  (60-n) × (C/P)³**.

**PRELOAD** — pressing the rolling elements against the raceways before any load is applied removes internal clearance and play. Done by shims, springs or matched angular-contact pairs; it raises stiffness and kills vibration and noise, at the cost of some friction and heat. Essential in machine-tool spindles, robot arms and anywhere positional accuracy matters. Where a shaft gets hot (a long drive between gears) and presses both rings, allow for thermal growth with a C3 clearance bearing.

## SECTION 9 · DIAGNOSIS

### BEARING FAILURES — RECOGNISE, CAUSE, PREVENT

A failing bearing tells its story in its own metal — read the damage and you find the cause. **Noise, roughness and heat are the first warnings**. Catch trouble early and you replace a cheap bearing instead of a whole shaft, gearbox or motor.

| Failure                   | What it looks like                                       | What causes it                                                                               | Prevent it                                             |
|---------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Spalling / pitting</b> | Small pits, chips or crescent flakes on raceway or balls | Fatigue after millions of cycles — normal end-of-life — or overload, misalignment, poor lube | Replace; check load, alignment and lubrication         |
| <b>Brinelling</b>         | Dents in the raceway at exactly the ball spacing         | Static overload — hammer blows, or pressing through the balls during mounting                | Never hammer; always press on the correct ring         |
| <b>False brinelling</b>   | Shiny fretting marks like a ghost of the balls           | Vibration while NOT rotating — transport, parked machines                                    | Rotate or preload during storage; not real damage      |
| <b>Overheating</b>        | Blue/brown/black rings; melted grease                    | Too much grease, over-preload, seal rub, too fast                                            | Pack 30–50%, check preload &amp; seals, use C3         |
| <b>Contamination</b>      | Dull, scratched, frosted raceways; gritty feel           | Dirt or particles in via bad seals / dirty mounting                                          | Sealed bearings, clean area, never blow clean with air |
| <b>Electrical fluting</b> | Washboard / corrugated lines on the raceway              | Stray current micro-arcs (VFD motors, shaft currents)                                        | Hybrid ceramic or grounding brushes                    |
| <b>Cage damage</b>        | Deformed, cracked or broken cage; loose balls            | Shock loads, mounting damage, too-fast, wrong lube                                           | Replace; cages are the weakest part — no shock         |
| <b>Corrosion / rust</b>   | Red-brown rust, pitting, rough running                   | Water, acid or condensation inside                                                           | Sealed bearings, fresh lube, store dry                 |

## SECTION 10 · QUICK REFERENCE

### COMMON BEARING SIZES — BORE × OD × WIDTH (INDICATIVE)

Real catalogue figures for the popular series (values are typical — always check the maker's data for your exact part). The pattern to notice: **62-series** is the light all-rounder, **63-series** is the heavier version of the same bore, **68/69-series** are thin-section for tight spaces, and the **60-series** splits the difference. For a given bore, the OD + width climb as the series gets heavier.

| Code     | Bore d | OD D  | Width B | Dynamic C | Static C0 | Typical home / school use            |
|----------|--------|-------|---------|-----------|-----------|--------------------------------------|
| 6001     | 12 mm  | 28 mm | 8 mm    | 5.1 kN    | 2.4 kN    | small motors, printers               |
| 608-2Z   | 8 mm   | 22 mm | 7 mm    | 3.35 kN   | 1.44 kN   | skateboards, scooters (the classic)  |
| 6201-2RS | 12 mm  | 32 mm | 10 mm   | 6.1 kN    | 3.0 kN    | wheels, RC cars, small shafts        |
| 6204     | 20 mm  | 47 mm | 14 mm   | 12.7 kN   | 6.6 kN    | pulley shafts, conveyor idlers       |
| 6205     | 25 mm  | 52 mm | 15 mm   | 14.0 kN   | 7.8 kN    | electric motors, fans                |
| 6305     | 25 mm  | 62 mm | 17 mm   | 20.7 kN   | 11.4 kN   | heavier motor / gearbox duty         |
| 6308     | 40 mm  | 90 mm | 23 mm   | 41.8 kN   | 24.0 kN   | pumps, blowers, gearboxes            |
| 6903     | 17 mm  | 30 mm | 7 mm    | 4.5 kN    | 2.5 kN    | thin, light — drones, RC             |
| 6804     | 20 mm  | 32 mm | 7 mm    | 4.0 kN    | 2.3 kN    | thin-section — mechatronics, pulleys |

## SECTION 11 · SLIDING TYPES

### PLAIN (SLEEVE) & LINEAR BEARINGS

Not every bearing rolls. **Plain (sleeve) bearings** let the shaft slide directly in a bush — simple, cheap, compact and quiet, ideal for oscillating or low-speed heavy duty.

**Plain bearing materials:**

- Bronze bushes** — classic, hard-wearing, good for oscillating loads
- Sintered / oil-impregnated** — porous bronze soaked in oil; self-lubricating for years
- PTFE-lined** — very low friction, dry running, chemical-resistant
- Plastic / polymer** — cheap, quiet, corrosion-proof, maintenance-free
- White metal (Babbitt)** — soft lining in big engine/crankshaft bearings

**LINEAR bearings move along a shaft instead of around it:**

- Linear ball bushing** — recirculating balls riding a round rail; smooth low-friction slide
- Linear rail / carriage** — ball or roller carriages on precision ground rails
- Dovetail / plain slides** — cheap sliding guides for light duty
- Pillow-block & split housings** — an insert bearing in a bolt-on cast housing (UC series) that clamps around a shaft; the classic for conveyor rollers and line shafting

Pillow blocks (suffix UC) are self-aligning ball bearings in a split housing with a grub screw or eccentric collar — they bolt straight to a frame and accept some misalignment.

## SECTION 12 · IN THE WILD

### APPLICATIONS — WHERE THEY RUN

One way to 'get' bearings is to see how the type follows the job:

| Machine                | Bearing it uses                  | Why                                          |
|------------------------|----------------------------------|----------------------------------------------|
| Electric motors        | Deep-groove ball 62/63           | Moderate load, high speed, cheap &amp; quiet |
| Gearboxes              | Cylindrical &amp; tapered roller | Heavy radial + axial shock                   |
| Vehicle wheels         | Tapered roller pair              | Heavy radial + axial, easy to adjust/preload |
| Bicycle / skate wheels | Deep-groove ball 608/6200        | Light, fast, cheap, sealed                   |
| Conveyors / idlers     | Pillow-block ball                | Bolt-on, self-aligning, easy to service      |
| Pumps &amp; blowers    | Deep-groove ball + angular       | Speed with moderate load                     |
| Fans                   | Deep-groove ball                 | Quiet high-speed running                     |
| Machine spindles       | Angular-contact pair / hybrid    | High speed, high precision, preload          |
| Pulleys &amp; belts    | Deep-groove ball                 | Radial belt tension                          |
| Car suspensions        | Deep-groove + needle             | Space, shock, sealed                         |

The golden rule everywhere: **match the bearing to the load, speed, space and contamination** — a sealed deep-groove ball covers 80% of machines; rollers and angular-contacts step in where loads and speeds climb.

## SECTION 13 · VOCABULARY

### GLOSSARY — 18 KEY TERMS

**Bore** — the shaft hole through the inner ring (d)

**Width** — the ring thickness along the shaft (B)

**Radial load** — force across the shaft, at right angles

**Clearance** — working gap between balls and races

**Race / raceway** — the hardened track the balls roll in

**Brinelling** — dents from static overload / hammering

**L10 life** — revs 90% of a batch will survive

**Static rating C0** — max load without permanent denting

**Snap ring** — retaining ring in a groove in the OD

**Seal (RS/2RS)** — contact rubber seal, greased for life

**Outer diameter** — the housing size, OD (D)

**Radial load** — force across the shaft, at right angles

**Clearance** — working gap between balls and races

**Race / raceway** — the hardened track the balls roll in

**Brinelling** — dents from static overload / hammering

**Dynamic rating C** — load for 10  revs at 90% survival

**Preload** — forcing elements against races to kill play

**Shield (2ZZ)** — non-contact metal cover

**Equivalent load P** — combined radial-axial value in the L10 formula

## EXAM TIP

### DT 0445 / F1-in-Schools QUICK RULE

Real balls = ball rings = 20 mm balls. Pick a ball ring that goes in. Never hammer. Pack grease 30–50%. Double load → 8× shorter life.